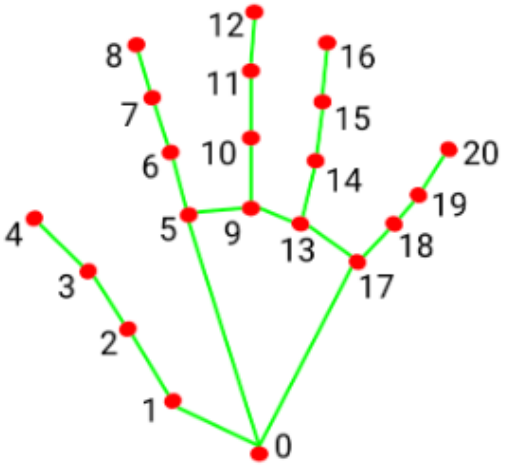
Hand detection

Hand detection
1. Hand detection
2. Preview of the effect
    2.1. The USB camera
    2.2. The CSI camera

MediaPipe is a cross-platform machine learning framework that can process the key point detection of hands, faces, postures and so on in videos and images in real time.

1. Hand detection

MediaPipe hand detection can identify and track the 21 key points of a hand.



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

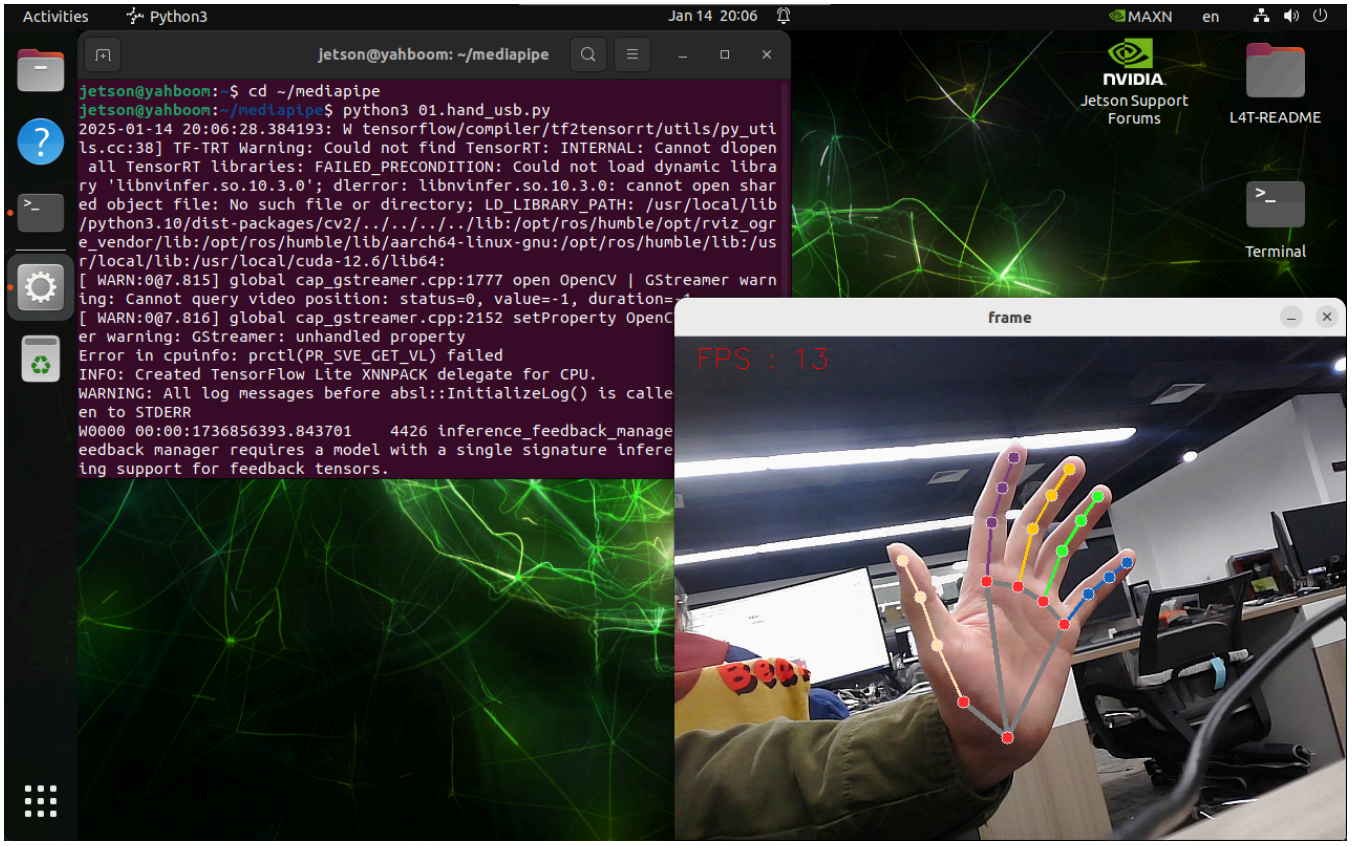
2. Preview of the effect

2.1. The USB camera

```
cd ~/mediapipe
```

```
python3 01.hand_usb.py
```

Note: click the camera window and press `q` to exit the program!

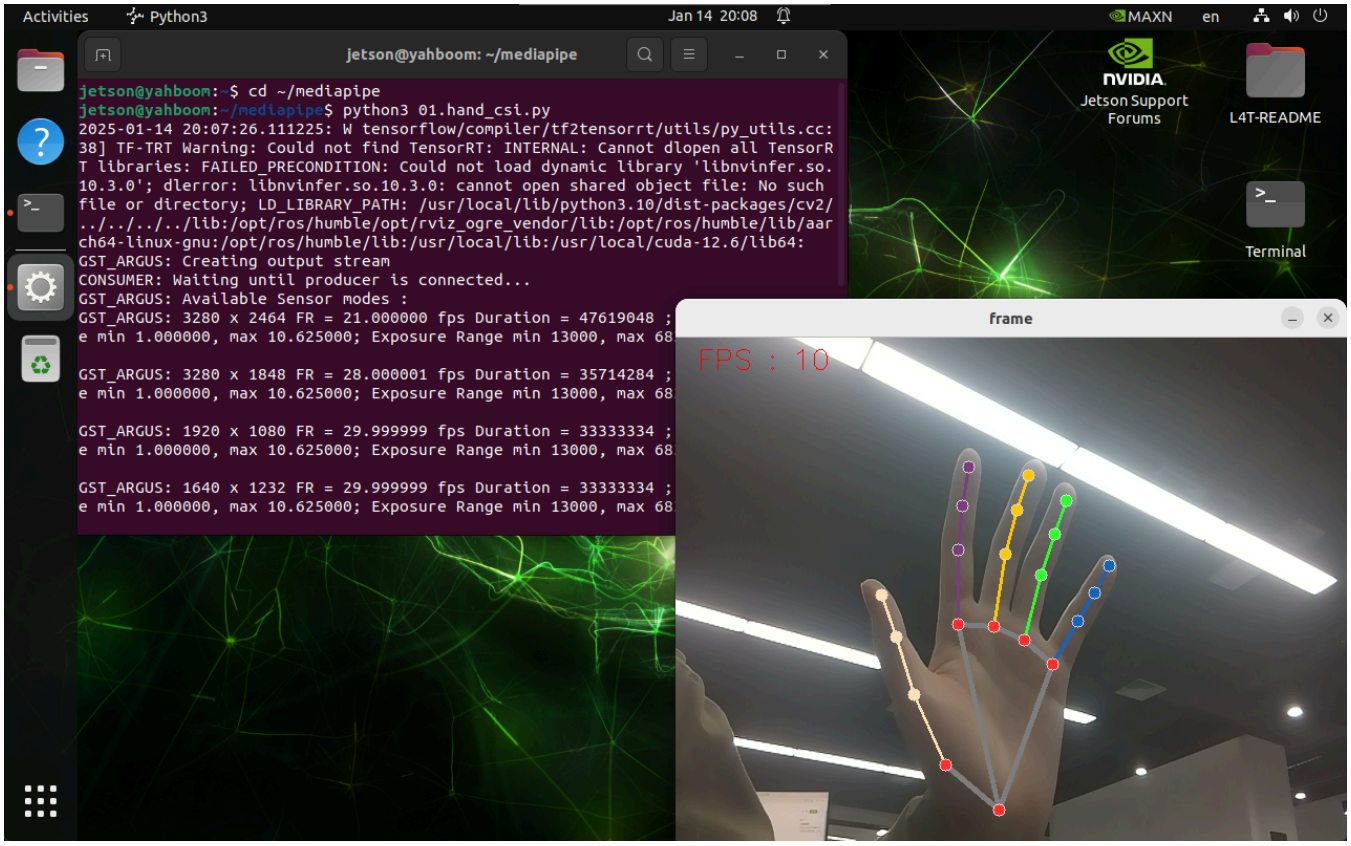


2.2. The CSI camera

```
cd ~/mediapipe
```

```
python3 01.hand_csi.py
```

Note: click the camera window and press `q` to exit the program!


Activities
Python3
Jan 14 20:08
MAXN
en
jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 01.hand_csi.py
2025-01-14 20:07:26.111225: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
NVIDIA
Jetson Support Forums
L4T-README
Terminal
frame
FPS : 10